

## TASK 6 – Formulate the Recurrence Relation

### Recursive Approach: Binary Search

#### 1. Input size of the original problem:

Let  $(n)$  be the total number of student records in the sorted database.

#### 2. Size of the recursive subproblem:

Binary Search divides the search range into two halves. After each comparison, the remaining search space is approximately  $(n/2)$ .

#### 3. Number of recursive calls:

At each step, only **one recursive call** is made, either to the left half or to the right half, depending on the comparison.

#### 4. Work performed outside the recursive call:

The algorithm compares the required registration number with the middle record and determines which half to search next. This requires constant time,  $(O(1))$ .

#### 5. Base case:

The recursion stops when the registration number is found at the middle position or when the search range becomes empty. The base case requires constant time,  $(T(1)=O(1))$ .

#### 6. Recurrence relation:

Since Binary Search reduces the problem size to half, makes one recursive call, and performs constant work outside the call, its recurrence relation is:

$$\begin{aligned} &[ \\ T(n) &= T\left(\frac{n}{2}\right) + O(1) \\ &] \end{aligned}$$

with the base case:

$$\begin{aligned} &[ \\ T(1) &= O(1) \\ &] \end{aligned}$$

Therefore, the time complexity of Recursive Binary Search is  **$O(\log n)$** .